Chapter 13. Ramsey Theory

Does It Clique?

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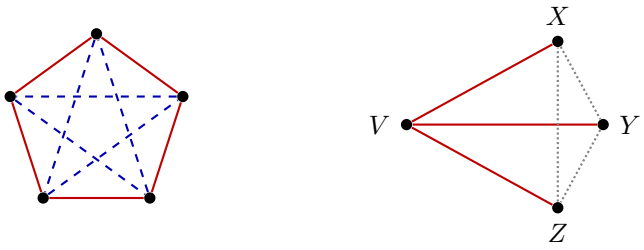
13.1 Ramsey theory for finite graphs

Motivation. Complete disorder is impossible: colour the edges of a large enough complete graph with two colours and a large monochromatic clique must appear. The proof is a double induction built on the recursion $R(k,\ell) \le R(k,\ell-1)+R(k-1,\ell)$; the exact values, however, are known only for very small cases.

Example 13.1

Among any six people there are three who all know each other or three who are mutual strangers.
Solution. Colour the edges of K_6 red (acquainted) or blue (strangers). A vertex V has degree 5, so by pigeon-hole at least three of its edges have the same colour, say red, going to X, Y, Z. If an edge of XYZ is red, it forms a red triangle with V; otherwise XYZ is a blue triangle.

Five people are not enough: colour the sides of a regular pentagon red and its diagonals blue. Every triangle contains both a side and a diagonal.



K_5: no monochromatic triangle Example 13.1

Figure. *Left: a 2-colouring of K_5 (solid red, dashed blue) without monochromatic triangles. Right: three red edges at V; any red edge of XYZ closes a red triangle.*

Equivalently: for every simple graph H on six vertices, H or its complement contains a **clique** (complete subgraph) of size 3.

Theorem 13.2 (Ramsey's theorem for graphs)

For integers $k,\ell \ge 2$ there is a least positive integer $R(k,\ell)$ such that every red/blue colouring of the edges of $K_{R(k,\ell)}$ contains a red K_k or a blue K_ℓ.

It suffices to find one such integer; the least one then exists.

Example 13.3

$R(3,3)=6$ by Example 13.1 and the pentagon. Trivially $R(2,2)=2$.

Proof of Theorem 13.2. Induction on $k+\ell$.

Step 1 (boundary). $R(k,2)=k$: either all edges of K_k are red, or one blue edge is a blue K_2. Likewise $R(2,\ell)=\ell$.

Step 2 (induction step). Assume $R(k,\ell-1)$ and $R(k-1,\ell)$ exist. We show

$$R(k,\ell) \le R(k,\ell-1)+R(k-1,\ell). \tag{13.1}$$

In K_N with $N=R(k,\ell-1)+R(k-1,\ell)$, a vertex V has $N-1$ edges, so it has at least $R(k,\ell-1)$ blue edges or at least $R(k-1,\ell)$ red edges.

(a) In the first case, the set b of blue neighbours contains a red K_k (done) or a blue $K_{\ell-1}$, which together with V is a blue K_ℓ.

(b) In the second case, the set r of red neighbours contains a blue K_ℓ or a red K_{k-1}, which together with V is a red K_k. ■

Corollary 13.4

For $k, \ell \geq 2$,

$$R(k,\ell) \leq R(k,\ell-1) + R(k-1,\ell). \tag{13.2}$$

Proof. This is (13.1), established in the proof of Theorem 13.2. ■

For instance $R(4,3) \leq R(4,2) + R(3,3) = 10$, which is not sharp:

Example 13.5

$R(4,3) = 9$.
Solution.
(1) K_9 *always works.* Some vertex V has (i) at least six red edges or (ii) at least four blue edges. Otherwise every vertex has exactly five red edges, and the red degree sum would be 45, which is odd.

(a) Six red neighbours span a red or blue triangle (Example 13.1); a blue triangle is done, and a red one plus V is a red K_4.

(b) Four blue neighbours: if all edges among them are red, they form a red K_4; a blue edge among them forms a blue triangle with V.

(2) K_8 *need not work.* On vertices $1, \ldots, 8$, colour $\{i,j\}$ $(i<j)$ blue if $j-i \in \{1,4,7\}$, red if $j-i \in \{2,3,5,6\}$. A blue triangle with least vertex i needs two of $i+1, i+4, i+7$, but their differences $3, 6, 3$ are red. A red K_4 with least vertex i needs three of $i+2, i+3, i+5, i+6$, which forces $i+2, i+3$ or $i+5, i+6$ together, whose difference 1 is blue.

Example 13.6

$R(4,4) = 18$.
Solution. The upper bound is $R(4,4) \leq R(4,3) + R(3,4) = 18$. For the lower bound, colour K_{17} on $\mathbb{Z}_{17}$: $\{i,j\}$ is red iff $i-j$ is a nonzero square mod 17, i.e. $i-j \in Q = \{\pm1, \pm2, \pm4, \pm8\}$ (well defined since $-1 \in Q$).
[Supplement] *No monochromatic* K_4. Translations $x \mapsto x+c$ and multiplications $x \mapsto qx$ with $q \in Q$ preserve the red graph. Suppose $\{0,1,a,b\}$ after normalisation is a red K_4 (translate one vertex to 0, then multiply by the inverse of another, which lies in Q). Then $a, b \in Q \cap (1+Q) \setminus \{0,1\} = \{2,9,16\}$, but the pairwise differences of $2, 9, 16$ are ±7 and ±14, which are not in Q. Hence there is no red K_4. Multiplication by the non-square 3 maps red edges to blue and blue to red, so there is no blue K_4 either.

$R(4,5) = 25$ and $43 \leq R(5,5) \leq 48$ (bounds as quoted in the book); $R(k,k)$ is unknown for $k \geq 5$.

Theorem 13.7

For $k, \ell \geq 2$,

$$R(k,\ell) \leq \binom{k+\ell-2}{k-1}. \tag{13.3}$$

Proof. Induction on $k+\ell$. For $k=2$ both sides equal ℓ, and symmetrically for $\ell=2$. Otherwise,

by Corollary 13.4, the induction hypothesis and Pascal's rule,

$$R(k,\ell) \le \binom{k+\ell-3}{k-1} + \binom{k+\ell-3}{k-2} = \binom{k+\ell-2}{k-1}.$$

■

Corollary 13.8

$R(k,k) \le 4^{k-1}$ for $k \ge 2$.

Proof. By Theorem 13.7, $R(k,k) \le \binom{2k-2}{k-1} \le \sum_i \binom{2k-2}{i} = 2^{2k-2}$. ■

Campos, Griffiths, Morris and Sahasrabudhe (2023) showed $R(k,k) < (4-\epsilon)^k$ for some $\epsilon > 0$. Exponential lower bounds come from the probabilistic method (Chapter 15).

13.2 Generalizations of the Ramsey theorem

Motivation. The same induction works with any number of colours and even when we colour r-element subsets instead of edges. The hypergraph version has a striking geometric consequence: among enough points in general position, some n form a convex polygon.

Example 13.9

If every edge of K_{17} is coloured with one of three colours, there is a monochromatic triangle. *Solution.* A vertex V has 16 edges, so at least six have the same colour, say green, with endpoint set g. A green edge inside g closes a green triangle with V. Otherwise g is 2-coloured on at least six vertices and contains a red or blue triangle (Example 13.1).

Theorem 13.10

For positive integers $n_1, \dots, n_k$ there is a least $N = R(n_1, \dots, n_k)$ such that every colouring of the edges of K_n, $n \ge N$, with colours $1, \dots, k$ has, for some i, a K_{n_i} all of whose edges have colour i.

Proof (completed [Supplement]). Induction on $n_1 + \cdots + n_k$. If some $n_i = 1$, then $N = 1$ works (a single vertex is a K_1 of every colour). Otherwise let $R_i = R(n_1, \dots, n_i - 1, \dots, n_k)$, which exist by induction, and

$$N = \sum_{i=1}^{k} (R_i - 1) + 2.$$

A vertex V of K_N has $N - 1 = \sum_i (R_i - 1) + 1$ edges, so by pigeon-hole some colour i occurs on at least R_i of them. Let S be the set of their other endpoints. By the definition of R_i, K_S contains a K_{n_j} of colour j for some $j \ne i$ (done), or a $K_{n_i - 1}$ of colour i, which together with V is a K_{n_i} of colour i. ■

Instead of edges we may colour the r-element vertex sets (K_r-subgraphs) of K_n; $r = 2$ is the edge case.

Theorem 13.11 (Hypergraph Ramsey theorem)

Colour each r-element subset of an n-set with one of k colours, and let $n_1, \dots, n_k$ be positive integers. There is a least $N = R_r(n_1, \dots, n_k)$ such that for $n \geq N$ there are an index i and an n_i-set all of whose r-subsets have colour i.

Proof [Supplement]. The book omits the proof. We use induction on r, and for fixed r on $n_1 + \cdots + n_k$.

Step 1 ($r = 1$). Colouring points: $N = \sum_i (n_i - 1) + 1$ works by pigeon-hole.

Step 2 (trivial cases). If some $n_i < r$, any n_i-set has no r-subsets at all, so it qualifies vacuously; take $N = n_i$.

Step 3 (induction step). Let $r \geq 2$, all $n_i \geq r$, and let $R_i = R_r(n_1, \dots, n_i - 1, \dots, n_k)$ (smaller sum) and

$$N = 1 + R_{r-1}(R_1, \dots, R_k)$$

(exists since $r - 1 < r$). Fix a point V of an N-set X, and colour each $(r-1)$-subset S of $X - V$ by the colour of $S \cup \{V\}$. By the choice of N there are i and $T \subseteq X - V$ with $|T| = R_i$ such that $S \cup \{V\}$ has colour i for every $(r-1)$-set $S \subseteq T$. Apply the definition of R_i to the original colouring on T. Either some n_j-subset of T ($j \neq i$) is monochromatic of colour j, and we are done, or some $(n_i - 1)$-set $U \subseteq T$ has all r-subsets of colour i. Then $U \cup \{V\}$ has all its r-subsets of colour i: those avoiding V lie in U, and those containing V are $S \cup \{V\}$ with $S \subseteq T$. ∎

Theorem 13.12 (Erdős–Szekeres)

For every n there is a least $ES(n)$ such that among any $N \geq ES(n)$ points in the plane, no three collinear, some n form a convex n-gon.

Proof. We show that $R_3(n, n)$ points suffice. Number the points $1, \dots, N$. For a triangle, list its vertices in increasing order; colour it red if that path turns clockwise and blue if counterclockwise. By Theorem 13.11 there are n points all of whose triangles have the same colour.

Step 1 (reduction) [Supplement]. n points (no three collinear) fail to be in convex position iff one of them lies inside a triangle formed by three others. Indeed, if a point P is not a vertex of the convex hull of the others, triangulate that hull by diagonals from one vertex; P lies in one of these triangles, and not on its boundary since no three points are collinear.

Step 2 (no point inside a triangle). Let $\chi(x, y, z) = \pm 1$ be the orientation of (x, y, z). It is invariant under cyclic shifts and changes sign under transpositions, and the colour of a triangle with $x < y < z$ is $\chi(x, y, z)$. Suppose D lies inside triangle ABC. Then D is on the same side of each side line as the opposite vertex, so

$$\chi(A, B, D) = \chi(B, C, D) = \chi(C, A, D) = \chi(A, B, C) =: \sigma.$$

By a cyclic shift we may take A smallest among A, B, C, and by a reflection (which reverses all orientations and keeps the configuration monochromatic) we may take $A < B < C$, so all four triangles have colour σ. Then:

(a) If $D < A$: $\chi(D, A, C) = \chi(A, C, D) = -\chi(C, A, D) = -\sigma$.

(b) If $A < D < B$: $\chi(A, D, B) = -\chi(A, B, D) = -\sigma$.

(c) If $B < D < C$: $\chi(B, D, C) = -\chi(B, C, D) = -\sigma$.

(d) If $C < D$: $\chi(A, C, D) = -\sigma$.

Each case gives a triangle of colour $-\sigma$, a contradiction. So the n points are in convex position. ∎

The original Erdős–Szekeres argument gives $ES(n) \leq \binom{2n-4}{n-2} + 1$. They also showed $ES(n) \geq 2^{n-2} + 1$ and conjectured equality, known for $n \leq 6$. Suk (2017) proved $ES(n) \leq 2^{n+o(n)}$.

13.3 Ramsey theory in geometry

Motivation. Colourings of the whole plane obey Ramsey-type laws too: some configuration with prescribed distances must be monochromatic. Equivalently, the unit-distance graph of the plane has chromatic number at least 4 (in fact at least 5).

Example 13.13

If every point of the plane is red or blue, some two points at unit distance have the same colour.
Solution. Two vertices of an equilateral triangle of side 1 share a colour.

Example 13.14

The same holds with three colours.
Solution. Suppose not. In any unit equilateral triangle ABC all colours differ. Reflect A across BC to get D, so that BCD is also unit equilateral. Then D differs from B and C, so D has the colour of A, and $|AD| = \sqrt{3}$. As ABC was arbitrary, any two points at distance $\sqrt{3}$ have the same colour. Hence the circle k of radius $\sqrt{3}$ about a red point is entirely red. But k contains two points at distance 1 (any chord length up to $2\sqrt{3}$ occurs), a contradiction.

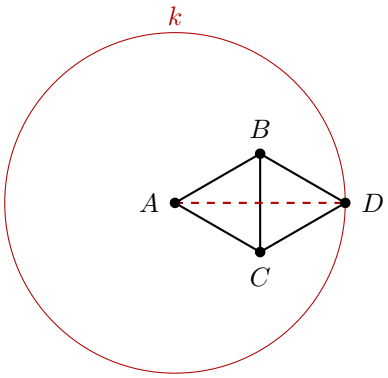


Figure. *Example 13.14: D is forced to have the colour of A, so every point of the circle k of radius $\sqrt{3}$ about A has that colour.*

Example 13.15

Colour the plane red/blue, and let T be a triangle with angles $30°, 60°, 90°$ and hypotenuse 1. Some triangle congruent to T has monochromatic vertices.
Solution. By Example 13.13 there is a unit segment AB with both ends red, say. On the circle with diameter AB, take $D_1, \dots, D_4$ dividing it, together with A, B, into six equal arcs. Any triangle ABD_i is congruent to T (Thales). If some D_i is red we are done. Otherwise all D_i are blue; two of them that are diametrically opposite form a diameter, and together with a third D_j they form a blue copy of T.

In graph language, the unit-distance graph of the plane has chromatic number (the **chromatic number of the plane**) greater than 3, by Example 13.14. It is at most 7, and de Grey (2018) proved it is at least 5; the exact value is open.

13.4 Selected exercises

Motivation. The book exercises below show the Ramsey mechanism (pigeon-hole at one vertex, then induction) in three settings: monotone subsequences, triangles in many colours, and monochromatic solutions of $a+b=c$.

Exercise 13.2 (Monotone subsequences)

Prove that every permutation of length $nm+1$ has an increasing subsequence of length $n+1$ or a decreasing subsequence of length $m+1$. (The entries of a subsequence need not be in consecutive positions.)

Solution.

Step 1 (labels). For each entry x let $\lambda(x)$ be the length of the longest increasing subsequence ending at x. If some $\lambda(x) \geq n+1$, we are done; so assume $\lambda(x) \in [n]$ for all x.

Step 2 (pigeon-hole). The $nm+1 > mn$ entries receive labels from $[n]$, so by Theorem 1.4 at least $m+1$ entries have the same label.

Step 3 (equal labels decrease). Let x precede y with $\lambda(x) = \lambda(y)$. If $x < y$, appending y to a longest increasing subsequence ending at x would give $\lambda(y) \geq \lambda(x)+1$. Hence $x > y$, and the entries with a common label, read from left to right, form a decreasing subsequence. By Step 2 it has length at least $m+1$.

The bound is sharp: for $n=2$, $m=2$ the permutation $2\,1\,4\,3$ of length 4 has neither an increasing nor a decreasing subsequence of length 3. ■

Exercise 13.6 (Monochromatic triangles in k colours)

Let $n_k = k!\left(1+\frac{1}{1!}+\frac{1}{2!}+\cdots+\frac{1}{k!}\right)+1$. Prove that every colouring of the edges of K_{n_k} with k colours contains a triangle with monochromatic edges.

Solution. Let $m_k = n_k - 1 = \sum_{j=0}^{k} \frac{k!}{j!}$, an integer. For $k \geq 2$,

$$m_k = \sum_{j=0}^{k-1} k \cdot \frac{(k-1)!}{j!} + \frac{k!}{k!} = k\, m_{k-1} + 1.$$

We prove by induction on $k \geq 1$ that every k-colouring of the edges of K_{m_k+1} has a monochromatic triangle.

Step 1 (initial step). $m_1 = 2$, and in a 1-colouring of K_3 the triangle itself is monochromatic.

Step 2 (pigeon-hole at a vertex). Let $k \geq 2$ and fix a vertex v of K_{m_k+1}. It has $m_k = k\, m_{k-1}+1$ edges, so by Theorem 1.4 some colour, say colour 1, appears on at least $m_{k-1}+1$ of them. Let S be the set of their other endpoints, $|S| \geq m_{k-1}+1$.

Step 3 (inside S). If some edge inside S has colour 1, it forms a triangle of colour 1 with v. Otherwise the edges inside S use only the $k-1$ colours $2,\dots,k$, and by the induction hypothesis (applied to $m_{k-1}+1$ vertices of S) there is a monochromatic triangle in S.

For $k=2$ this gives $n_2 = 6$, i.e. $R(3,3) \leq 6$, and for $k=3$ it gives $n_3 = 17$, matching Example 13.9. ■

Exercise 13.16 (Schur's theorem)

Colour every positive integer with one of k colours. Prove that there is an integer $N = N(k)$ such that for every $n > N$ there are integers $a, b, c < n$ of the same colour with $a + b = c$ (with $a = b$ allowed).

Solution.

Step 1 (the constant). Let $N = R(3, 3, \ldots, 3)$ with k entries 3, which exists by Theorem 13.10. Let $n > N$.

Step 2 (an edge colouring). Take the complete graph on the vertices $0, 1, \ldots, N-1$, and give the edge $\{i, j\}$ the colour of the positive integer $|i - j| \in [N-1]$.

Step 3 (a monochromatic triangle). By the choice of N there are vertices $i < j < l$ whose three edges have the same colour. Put $a = j - i$, $b = l - j$ and $c = l - i$. These have the same colour, $a + b = c$, and $1 \le a, b, c \le N - 1 < n$. ■